

WebTrap

Active Defense Honeypot & Threat Profiler

Final Year Cybersecurity Project

1. The Threat Landscape

- Modern cyber attacks are **highly automated** and continuous.
- Hackers deploy massive botnets to scan thousands of web servers per second.
- Traditional defenses (Firewalls/IDS) rely on passive blocking.
- This creates a massive volume of alerts, leading to **Alert Fatigue**.

2. Passive vs. Active Defense

Traditional Firewalls

Simply drop the connection when a threat is detected.

Result: Zero intelligence is gathered.

Active Defense

Engages the attacker, tricks them, and studies them.

Result: Gathers Intelligence & wastes attacker resources.

3. Proposed Solution: WebTrap

A custom Deception Technology framework.

- **The Lure:** A simulated Enterprise Login Gateway.
- **The Trap:** Silently logs credential stuffing and payloads.
- **The Profiler:** Automatically traces the hacker's ISP.
- **The Retaliation:** Wastes attacker's computing resources.

4. System Architecture

Built using **Python 3** and **Bash** for zero-dependency deployment.



Python Engine

- Handles HTTP Requests.
- Executes Deep Packet Inspection.
- Applies Tarpit Defense.



Bash Profiling Engine

- Asynchronous Daemon.
- OSINT API Integration.
- Generates HTML Dashboards.

5. Deep Packet Inspection (DPI)

Analyzes HTTP payloads against a custom Regex Engine.

```
# Threat Signatures Engine
SIGNATURES = {
    "SQL_INJECTION": re.compile(r"(%27)|(')|(--)|(UNION)|
(SELECT)"),
    "XSS_SCRIPTING": re.compile(r"(
```